BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 313 · Processors, scheduling and LMC practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does LMC STA do?

Q2. What is the role of MAR?

Q3. What is a time quantum?

Q4. Full adder 1+1+0: sum and carry bits?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

ACC — Accumulator register.

FCFS — First Come First Served.

SJF — Shortest Job First.

pipeline — Overlap instruction stages.

New learning

Understand — Trace the LMC accumulator and mailbox separately: INP overwrites ACC, while STA copies to storage.

Apply — Schedule independent CPU jobs with an explicit arrival/time rule and label every Gantt boundary.

Evaluate — Distinguish processor speed, memory cache and interrupt/OS services when explaining performance.

Worked example

Trace sequential instruction fetch when PC=90.

1. Predict first — PC holds address of next instruction, not the instruction contents.

2. Explain and check — Explain the reason for the result.

We Do · Practise with guidance

FCFS all tasks t0: A=3,B=2. Show timeline and mean waiting time.

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A 0–3,B 3–5; waits A0,B3; mean1.5.
```

Predict first

Arrival-order service makes B wait three CPU units.

Explain and check

Explain the reason for the result.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Using LMC code above, trace ACC after each input, LDA, SUB and OUT. [6]

Question 2

Processes A (burst 4), B (burst 2) and C (burst 1) all arrive at t=0. For FCFS, use the stated arrival-list order A, B, C to break ties; for non-preemptive SJF choose the shortest available job. Compare execution orders and mean waiting times. [6]

Question 3

Evaluate clock speed, cache and pipelining for a task with heavy memory latency and branches. [6]

Exit Ticket · Answer before checking

LMC STA X loads ACC from mailbox X.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Copies ACC contents to the specified mailbox.

Q2. Address of the memory location being accessed.

Q3. Maximum CPU slice before RR rescheduling.

Q4. 10_2: sum0, carry1.

You Do pages · Independent questions

Q1. ACC4 [1], A4 [1], ACC2 [1], B2 [1], LDA A→4 [1], SUB B→2 and OUT2 [1].

Q2. FCFS A,B,C waits 0,4,6 mean10/3 [3]; SJF C,B,A waits 0,1,3 mean4/3 [3].

Q3. Clock frequency matters [1], cache misses stall [1], cache may reduce latency [1], pipeline overlaps stages [1], branches can flush/stall [1], performance depends on bottlenecks [1].

Exit Ticket · Model answer

False. LDA X loads; STA X stores.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: